

SAT Math Mastery · Week 1A

Linear Equations: Structure, Rearranging & Translating

Time 150 minutes: 110 min solving + 40 min self-check and error log
Topics solution count (one / none / infinite) · structure before solving · solving for an expression · rearranging formulas · translating words · identities · linear inequalities in one and two variables

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Solve any linear equation (distribute, clear fractions with the LCD, collect, divide).
S2. Look at the *structure* first: spot a repeated block and solve for the block or for a target expression.
S3. Decide *one / none / infinitely many* solutions by comparing coefficients, including with parameters.
S4. Match coefficients when two expressions are equal for every x .
S5. Rearrange formulas (make any letter the subject, even when it appears twice).
S6. Translate words: “less than”, “at least”, consecutive integers and multiples, fixed fee plus rate, ages, mixtures.
S7. Linear inequalities: sign reversal, compound inequalities, integer counts, rounding to a feasible whole number.
S8. Two-variable inequalities: test points, read a shaded half-plane or region, count lattice points.

2 · REQUIRED FACTS

- Equal operations on both sides keep an equation true; multiplying or dividing an inequality by a negative number reverses it.
- “ n less than m ” means $m - n$; “at least” is $\geq$; “at most” and “no more than” are $\leq$.
- Consecutive integers differ by 1; consecutive even or odd integers differ by 2; consecutive multiples of 3 differ by 3.
- Every linear equation reduces to $mx = n$: one solution if $m \neq 0$, none if $m = 0$, $n \neq 0$, infinitely many if $m = n = 0$.
- In a graph of an inequality, a dashed boundary means $<$ or $>$ and a solid boundary means $\leq$ or $\geq$; the shading is the side that makes the inequality true.
- A count of whole objects must be a whole number: round *up* for “at least”, *down* for “at most”.

3 · REQUIRED FORMULAS

$$ax + b = cx + d \implies (a - c)x = d - b, \quad x = \frac{d - b}{a - c} \quad (a \neq c)$$

$$\text{total} = \text{fixed fee} + \text{rate} \times \text{quantity}, \quad \text{mixture: amount of solute} = \text{concentration} \times \text{volume}$$

$$n, n + 2, n + 4 \quad (\text{consecutive odd or even}), \quad \text{sum of } k \text{ terms} = k \cdot (\text{mean}), \quad ax + by \leq c \quad (\text{half-plane})$$

4 · MINI THEORY SHEET

Master reduction. Every linear equation collapses to $mx = n$:

$m \neq 0$	one solution	$x = \frac{n}{m}$
$m = 0, n \neq 0$	no solution	$0 = n$ is false
$m = 0, n = 0$	infinitely many	$0 = 0$ is true for every x

Structure before solving. Look for a block that repeats or a factor you can pull out. If $cE = k$ then $E = \frac{k}{c}$; you never need x itself when the question asks for E .

Identity in x . If two expressions are equal for every x , the x -coefficients match *and* the constants match. A constant that multiplies another constant (cd) is still just a constant.

Rearranging. Collect every term containing the target letter on one side, factor it out, then divide. Dividing by a bracket that contains letters is allowed when it is nonzero.

Inequalities. Same steps as equations; reverse only on multiplying or dividing by a negative. For a compound inequality, do the same operation to all three parts. A solution set written “ $x \leq c$ ” tells you the sign of the coefficient you divided by.

Contexts. Fixed + rate: $y = \text{fee} + \text{rate} \cdot n$. For “at most” counts solve then round down; for “at least” round up, and always check the boundary values.

Two-variable inequalities. Draw the boundary line, dashed for strict, solid otherwise. Test one point not on the line; a system is the overlap of the shaded regions.

Speed shortcuts.

Situation	Shortcut
asked for $x + y$ or $2x - y$	add or subtract the equations; never solve for x and y
asked for a block such as $2x - 5$	factor the given expression to show the block
none / infinite with a parameter	set the x -coefficients equal, then compare the constants
true for all x	plug in $x = 0$ and $x = 1$ to get two equations
“at least k less than E ”	$\geq E - k$ (the k is subtracted from E)
solution set given, find a constant	the sign of the divisor decides the direction of the inequality
“could NOT be” with ranges	find the open or closed interval of values; endpoints of strict ranges are excluded
which region / which inequality	test one point per boundary; dashed means strict
whole-number answer	solve, round the correct way, then verify both neighbours

5 · GUIDED EXAMPLES

Example 1 — Structure first: solution count with a parameter

For what value of a does $3(ax - 2) = 12x - 6$ have infinitely many solutions?

Expand: $3ax - 6 = 12x - 6$. The constants already agree, so the x -coefficients must agree: $3a = 12$, so $a = 4$.

Example 2 — Solve for the expression, not the variable

If $4a + 3b = 18$ and $a + 2b = 7$, what is the value of $a + b$?

Add the equations: $5a + 5b = 25$, so $a + b = 5$. (Check: $a = 3$, $b = 2$.)

Example 3 — Words → inequality → whole number

A rental costs a \$35 fee plus \$12 per hour. Maria can spend at most \$100. What is the greatest whole number of hours she can rent?

$35 + 12h \leq 100 \Rightarrow h \leq \frac{65}{12} \approx 5.42$. Hours must be whole, so round down: $h = 5$ (cost \$95; 6 hours would cost \$107).

6 · PRACTICE SET — 40 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

If $6x - 15 = 33$, what is the value of $2x - 5$?

(A) 8

(B) 11 ✓

(C) 16

(D) 33

- Factor the left side: $6x - 15 = 3(2x - 5)$.
- So $3(2x - 5) = 33$, and dividing by 3 gives $2x - 5 = 11$.
- (Check: $x = 8$ and $2(8) - 5 = 11$.)

Answer: (B)

2.

EASY · NO CALCULATOR NEEDED

The formula $C = \frac{5}{9}(F - 32)$ converts a temperature of F degrees Fahrenheit to C degrees Celsius. Which of the following gives F in terms of C ?

(A) $F = \frac{5}{9}C + 32$ (B) $F = \frac{9}{5}C - 32$ (C) $F = \frac{9}{5}C + 32$ ✓(D) $F = \frac{9}{5}(C - 32)$

- Multiply both sides by $\frac{9}{5}$: $\frac{9}{5}C = F - 32$.
- Add 32 to both sides: $F = \frac{9}{5}C + 32$.

Answer: (C)

3.

EASY · CALCULATOR

A mobile data plan costs a one-time activation fee of \$24 plus \$7.50 for each gigabyte (GB) of data used. The total cost for one customer was \$69. How many GB of data did the customer use?

- Write the cost: $24 + 7.5g = 69$.
- Subtract 24: $7.5g = 45$.
- Divide by 7.5: $g = 6$.

Answer: 6

4.

EASY · NO CALCULATOR NEEDED

Which of the following values of x satisfies the inequality $3 - 2x > 11$?

(A) -6 ✓

(B) -4

(C) 0

(D) 4

- Subtract 3: $-2x > 8$.
- Divide by -2 and *reverse* the inequality: $x < -4$.
- Only -6 is less than -4 . ($x = -4$ gives $11 > 11$, which is false.)

Answer: (A)

5.

EASY · NO CALCULATOR NEEDED

How many solutions does the equation $3(2x + 4) = 6x + 7$ have?

(A) Exactly two solutions

(B) Infinitely many solutions

(C) Exactly one solution

(D) No solution ✓1. Distribute: $6x + 12 = 6x + 7$.2. Subtract $6x$ from both sides: $12 = 7$, which is false for every x .

3. The equation has no solution.

Answer: (D)

6.

MEDIUM · NO CALCULATOR NEEDED

The sum of four consecutive odd integers is 112. What is the greatest of the four integers?

(A) 27

(B) 29

(C) 31 ✓

(D) 25

1. Let the integers be n , $n + 2$, $n + 4$, $n + 6$.2. Sum: $4n + 12 = 112$, so $4n = 100$ and $n = 25$.

3. The integers are 25, 27, 29, 31 (sum 112). The greatest is 31.

Answer: (C)

7.

MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\frac{2x - 1}{3} - \frac{x + 2}{4} = 1$?

1. The LCD is 12. Multiply every term by 12: $4(2x - 1) - 3(x + 2) = 12$.2. Distribute: $8x - 4 - 3x - 6 = 12$, so $5x - 10 = 12$.3. $5x = 22$, so $x = \frac{22}{5} = 4.4$.**Answer: $\frac{22}{5}$**

8.

MEDIUM · NO CALCULATOR NEEDED

$$5x + 3y = 29$$

$$3x + 5y = 19$$

The solution to the system of equations above is (x, y) . What is the value of $x + y$?

(A) 3

(B) 6 ✓

(C) 24

(D) 48

1. Add the two equations: $8x + 8y = 48$.

2. Divide by 8: $x + y = 6$.

3. (No need to find $x = \frac{11}{2}$ and $y = \frac{1}{2}$ separately.)

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

The sum S of the first n terms of an arithmetic sequence with first term a and last term ℓ is given by $S = \frac{n}{2}(a + \ell)$. Which of the following expresses a in terms of S , n and ℓ ?

(A) $a = \frac{2S - \ell}{n}$

(B) $a = \frac{S}{2n} - \ell$

(C) $a = \frac{2S}{n} + \ell$

(D) $a = \frac{2S}{n} - \ell$ ✓

1. Multiply both sides by $\frac{2}{n}$: $\frac{2S}{n} = a + \ell$.

2. Subtract ℓ : $a = \frac{2S}{n} - \ell$.

Answer: (D)

10.

MEDIUM · NO CALCULATOR NEEDED

$$(k - 3)x + 8 = 2x + 3k - 1$$

In the equation above, k is a constant. For what value of k does the equation have no solution?

1. Collect the x -terms on the left and constants on the right: $(k - 3)x - 2x = 3k - 1 - 8$, so $(k - 5)x = 3k - 9$.

2. No solution requires a zero coefficient with a nonzero right side: $k - 5 = 0$, so $k = 5$.

3. Check: at $k = 5$ the right side is $3(5) - 9 = 6 \neq 0$, so the equation reads $0 = 6$, which has no solution.

Answer: 5

11.

MEDIUM · NO CALCULATOR NEEDED

For a fundraiser, the number of tickets t sold must be at least 15 less than three times the number of volunteers v . Which inequality represents this requirement?

(A) $t \geq 3v - 15$ ✓

(B) $t \leq 3v - 15$

(C) $t \geq 3(v - 15)$

(D) $t \geq 15 - 3v$

1. “Three times the number of volunteers” is $3v$.
2. “15 less than” that is $3v - 15$.
3. “At least” means t is greater than or equal to this: $t \geq 3v - 15$.

Answer: (A)

12.

MEDIUM · CALCULATOR

The table shows three values of the linear function g . For what value of x is $g(x) = -10$?

x	3	7	11
$g(x)$	14	30	46

(A) -2

(B) -3 ✓

(C) $-\frac{5}{2}$

(D) 3

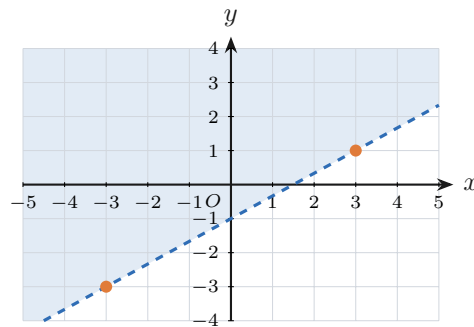
1. Slope: $\frac{30 - 14}{7 - 3} = 4$ (and $\frac{46 - 30}{11 - 7} = 4$, so the table is linear).
2. $14 = 4(3) + b \Rightarrow b = 2$, so $g(x) = 4x + 2$.
3. $4x + 2 = -10 \Rightarrow 4x = -12 \Rightarrow x = -3$.

Answer: (B)

13.

MEDIUM · DESMOS

The shaded region in the xy -plane shows the solution set of an inequality. The boundary line is dashed. Which of the following inequalities defines the shaded region?



(A) $y \geq \frac{2}{3}x - 1$

(B) $y > \frac{3}{2}x - 1$

(C) $y > \frac{2}{3}x - 1$ ✓

(D) $y < \frac{2}{3}x - 1$

- The dashed line passes through grid points $(-3, -3)$ and $(3, 1)$: slope $= \frac{1 - (-3)}{3 - (-3)} = \frac{2}{3}$.
- At $x = 0$ the line crosses at $y = -1$, so the boundary is $y = \frac{2}{3}x - 1$.
- A dashed boundary means a strict inequality, and the shading is above the line, so $y > \frac{2}{3}x - 1$.

Answer: (C)

14.

HARD · NO CALCULATOR NEEDED

$$4(ax + 3) - 2(x - b) = 18x + 20$$

In the equation above, a and b are constants. If the equation is true for all values of x , what is the value of ab ?

- Expand the left side: $4ax + 12 - 2x + 2b = (4a - 2)x + (12 + 2b)$.
- Match the x -coefficients: $4a - 2 = 18$, so $a = 5$.
- Match the constants: $12 + 2b = 20$, so $b = 4$.
- $ab = 5 \cdot 4 = 20$.

Answer: 20

15.

HARD · NO CALCULATOR NEEDED

$$3(kx - 2) = 12x + m$$

In the equation above, k and m are constants. Consider the following statements.

I. If $k = 4$ and $m = -6$, the equation has infinitely many solutions.

II. If $k \neq 4$, the equation has exactly one solution for every value of m .

III. If $m \neq -6$, the equation has no solution for every value of k .

Which of the statements must be true?

(A) I and II only ✓

(B) I, II, and III

(C) I only

(D) II only

1. Expand and collect: $3kx - 6 = 12x + m \Rightarrow (3k - 12)x = m + 6$.

2. I: with $k = 4$ and $m = -6$ the equation is $0 \cdot x = 0$, true for every x . *True*.

3. II: if $k \neq 4$ then $3k - 12 \neq 0$, so $x = \frac{m + 6}{3k - 12}$ is the one solution, whatever m is. *True*.

4. III: take $k = 1$, $m = 0$: $-9x = 6$ has the solution $x = -\frac{2}{3}$. So $m \neq -6$ does not force no solution. *False*.

5. Statements I and II only.

Answer: (A)

16.

HARD · NO CALCULATOR NEEDED

If $\frac{5}{3}(6w - 9) + 2 = 27$, what is the value of $2w - 3$?

(A) 8

(B) 25

(C) 4

(D) 5 ✓

1. Subtract 2: $\frac{5}{3}(6w - 9) = 25$.

2. Notice $6w - 9 = 3(2w - 3)$, so $\frac{5}{3} \cdot 3(2w - 3) = 5(2w - 3)$.

3. $5(2w - 3) = 25 \Rightarrow 2w - 3 = 5$.

4. (Check: $w = 4$.)

Answer: (D)

17.

HARD · NO CALCULATOR NEEDED

$$3(p - 2q) = 4(r - p)$$

Which of the following expresses p in terms of q and r ?

(A) $p = \frac{6q - 4r}{7}$

(B) $p = \frac{4r + 6q}{5}$

(C) $p = \frac{4r + 6q}{7}$ ✓

(D) $p = \frac{4r - 6q}{7}$

1. Distribute: $3p - 6q = 4r - 4p$.
2. Collect the p terms on the left: $3p + 4p = 4r + 6q$, so $7p = 4r + 6q$.
3. Divide by 7: $p = \frac{4r + 6q}{7}$.

Answer: (C)

18.

HARD · NO CALCULATOR NEEDED

What is the sum of all integers x that satisfy $-5 \leq \frac{4 - 3x}{2} < 8$?

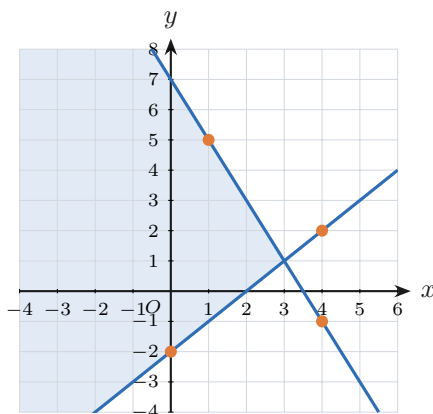
1. Multiply all three parts by 2: $-10 \leq 4 - 3x < 16$.
2. Subtract 4: $-14 \leq -3x < 12$.
3. Divide by -3 and reverse both inequalities: $-4 < x \leq \frac{14}{3}$.
4. Integers: $-3, -2, -1, 0, 1, 2, 3, 4$ (note -4 is excluded and $\frac{14}{3} \approx 4.67$).
5. Sum = $(-3 + 3) + (-2 + 2) + (-1 + 1) + 0 + 4 = 4$.

Answer: 4

19.

HARD · DESMOS

The shaded region in the xy -plane is the solution set of a system of two linear inequalities, and both boundary lines are solid. Which of the following systems has this solution set?



(A) $\begin{cases} y \geq -2x + 7 \\ y \leq x - 2 \end{cases}$

(B) $\begin{cases} y \geq x - 2 \\ y \leq -2x + 7 \end{cases}$ ✓

(C) $\begin{cases} y \leq x - 2 \\ y \leq -2x + 7 \end{cases}$

(D) $\begin{cases} y \geq x - 2 \\ y \geq -2x + 7 \end{cases}$

- Boundary 1 passes through $(0, -2)$ and $(4, 2)$: slope 1, so $y = x - 2$. The shading is on the upper-left side, above this line: $y \geq x - 2$.
- Boundary 2 passes through $(1, 5)$ and $(4, -1)$: slope -2 , so $y = -2x + 7$. The shading is below this line: $y \leq -2x + 7$.
- Check with the point $(-2, 2)$ inside the region: $2 \geq -4$ and $2 \leq 11$. Both hold.

Answer: (B)

20.

HARD · CALCULATOR

A club sells T-shirts for \$18 each, and each T-shirt costs the club \$7 to make. It sells mugs for \$12 each, and each mug costs the club \$5 to make. The club also pays a fixed booth fee of \$150. The club sells 27 T-shirts. What is the least whole number of mugs it must sell to make a profit of at least \$800?

- Profit per T-shirt: $18 - 7 = 11$; profit per mug: $12 - 5 = 7$.
- Profit: $27(11) + 7m - 150 \geq 800$, so $297 + 7m - 150 \geq 800$.
- $7m \geq 653 \Rightarrow m \geq \frac{653}{7} \approx 93.29$.
- Round up: $m = 94$. (Check: 93 mugs give profit $147 + 651 = 798 < 800$; 94 give 805.)

Answer: 94

21.

HARD · NO CALCULATOR NEEDED

If $1 < x < 6$ and $-2 < y < 3$, which of the following could *not* be the value of $2x - 3y$?

(A) -6.5 (B) 0 (C) 17.5 (D) -7 ✓

1. Multiply $1 < x < 6$ by 2: $2 < 2x < 12$.

2. Multiply $-2 < y < 3$ by -3 and reverse: $-9 < -3y < 6$.

3. Add: $-7 < 2x - 3y < 18$, with both endpoints excluded.

4. -7 is the lower endpoint and cannot be reached; -6.5 , 0 and 17.5 all lie inside the interval.

Answer: (D)

22.

HARD · NO CALCULATOR NEEDED

Four consecutive multiples of 3 are listed in increasing order. Three times the sum of the two smallest of them equals the largest increased by 60. What is the largest of the four numbers?

1. Let the numbers be m , $m + 3$, $m + 6$, $m + 9$.

2. The two smallest sum to $2m + 3$, so $3(2m + 3) = (m + 9) + 60$.

3. $6m + 9 = m + 69 \Rightarrow 5m = 60 \Rightarrow m = 12$.

4. The numbers are 12, 15, 18, 21. Check: $3(27) = 81 = 21 + 60$. The largest is 21.

Answer: 21

23.

HARD · NO CALCULATOR NEEDED

The linear functions f and g are defined by the tables below. For one value of t , the equation $f(x) = g(x)$ has no solution. What is that value of t ?

x	0	2
$f(x)$	5	11

x	1	3
$g(x)$	4	t

(A) 10 ✓

(B) 13

(C) 16

(D) 7

- Slope of f : $\frac{11-5}{2-0} = 3$, and $f(0) = 5$, so $f(x) = 3x + 5$.
- $f(x) = g(x)$ has no solution only when the graphs are parallel and distinct, so g must also have slope 3.
- $\frac{t-4}{3-1} = 3 \Rightarrow t = 10$.
- Then $g(x) = 3x + 1$, which has a different intercept from f , so the lines are parallel and distinct. Any other t gives a different slope and exactly one solution.

Answer: (A)**24.**

SAT CHALLENGE · NO CALCULATOR NEEDED

$$c(2x - 5) + d = 8x + cd$$

In the equation above, c and d are constants. If the equation is true for all values of x , what is the value of d ?

- Expand the left side: $2cx - 5c + d$. The right side is $8x + cd$.
- Match the x -coefficients: $2c = 8$, so $c = 4$.
- Match the constants (the term cd is a constant): $-5c + d = cd$. With $c = 4$: $-20 + d = 4d$.
- $-20 = 3d$, so $d = -\frac{20}{3}$.

Answer: $-\frac{20}{3}$

25.

SAT CHALLENGE · CALCULATOR

$$(k + 2)x = 36$$

In the equation above, k is an integer. For how many values of k is the solution x a positive integer less than 10?

1. If $k = -2$ the equation reads $0 = 36$ (no solution), so $k \neq -2$ and $x = \frac{36}{k+2}$.
2. x must be a positive integer below 10 that makes $k + 2 = \frac{36}{x}$ an integer: x is a divisor of 36 less than 10.
3. Divisors of 36 below 10: 1, 2, 3, 4, 6, 9 (not 5, 7, 8).
4. These give $k + 2 = 36, 18, 12, 9, 6, 4$, i.e. $k = 34, 16, 10, 7, 4, 2$.
5. There are 6 values of k .

Answer: 6

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$a(x - 4) \geq 6 - 2x$$

In the inequality above, a is a constant. The solution set of the inequality is all real numbers x such that $x \leq \frac{9}{2}$. What is the value of a ?

1. Expand and collect: $ax - 4a \geq 6 - 2x \Rightarrow (a + 2)x \geq 6 + 4a$.
2. The given solution set is $x \leq \dots$, but the inequality has $\geq$. So dividing by $(a + 2)$ must have reversed it: $a + 2 < 0$.
3. Then $x \leq \frac{6 + 4a}{a + 2}$. Set this equal to $\frac{9}{2}$: $2(6 + 4a) = 9(a + 2)$, so $12 + 8a = 9a + 18$ and $a = -6$.
4. Check: $a + 2 = -4 < 0$ ✓, and $-4x \geq -18 \Rightarrow x \leq \frac{9}{2}$.

Answer: -6

27.

SAT CHALLENGE · DESMOS

In the xy -plane, how many points (x, y) with positive integer coordinates satisfy both $3x + 2y \leq 17$ and $x - y \geq -2$?

1. Rewrite the second inequality as $y \leq x + 2$ and the first as $y \leq \frac{17 - 3x}{2}$.
2. $x = 1$: $y \leq \min(7, 3) = 3$, so 3 points. $x = 2$: $y \leq \min(5, 4) = 4$, so 4 points.
3. $x = 3$: $y \leq \min(4, 5) = 4$, so 4 points. $x = 4$: $y \leq \min(2, 6) = 2$, so 2 points.
4. $x = 5$: $y \leq \min(1, 7) = 1$, so 1 point. $x = 6$: $3(6) + 2 = 20 > 17$, so none (larger x fail too).
5. Total: $3 + 4 + 4 + 2 + 1 = 14$.

Answer: 14**28.**

SAT CHALLENGE · CALCULATOR

A tank holds 120 liters of a salt solution that is 30% salt by volume. Some liters of the solution are drained from the tank and replaced with the same number of liters of a solution that is 80% salt, so that the tank again holds 120 liters. The resulting solution is 50% salt. How many liters were drained?

1. Let r be the liters drained. Salt initially: $0.30(120) = 36$ liters.
2. Salt removed: $0.30r$. Salt added: $0.80r$.
3. New salt amount = $0.50(120) = 60$, so $36 - 0.30r + 0.80r = 60$.
4. $0.50r = 24 \Rightarrow r = 48$.

Answer: 48

29.

SAT CHALLENGE · CALCULATOR

For each integer t , let P_t be the point $(t + 1, 2t - 3)$ in the xy -plane. For how many integer values of t does P_t lie in the solution set of the system $3x - 4y > 10$ and $x + 3y \geq -22$?

(A) 2

(B) 3 ✓

(C) 4

(D) 5

1. Substitute $x = t + 1$, $y = 2t - 3$ into the first inequality: $3(t + 1) - 4(2t - 3) = 15 - 5t > 10$, so $t < 1$.
2. Second inequality: $(t + 1) + 3(2t - 3) = 7t - 8 \geq -22$, so $t \geq -2$.
3. Integers with $-2 \leq t < 1$: $t = -2, -1, 0$. ($t = 1$ gives $10 > 10$, which is false; $t = -2$ gives $-22 \geq -22$, which is true.)
4. There are 3 values of t .

Answer: (B)

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$k^2x - 4k = 9x - 12$$

In the equation above, k is a constant. Which of the following statements is true?

(A) $k = 3$: infinitely many; others: one(B) No solution for both $k = 3$ and $k = -3$ **(C) $k = 3$: infinitely many; $k = -3$: none ✓**(D) $k = -3$: infinitely many; $k = 3$: none

1. Collect the x -terms: $k^2x - 9x = 4k - 12$, so $(k^2 - 9)x = 4(k - 3)$.
2. $k = 3$: the equation is $0 \cdot x = 0$, true for every x , so infinitely many solutions.
3. $k = -3$: the equation is $0 \cdot x = -24$, false for every x , so no solution.
4. Every other k gives $k^2 - 9 \neq 0$ and exactly one solution $x = \frac{4}{k + 3}$. The first option is the true one.

Answer: (C)

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$k(x + 3) - 5x = 3x + 2m$$

In the equation above, k and m are constants. If the equation has infinitely many solutions, what is the value of $m - k$?

(A) 4 ✓

(B) 8

(C) 12

(D) 20

1. Expand and collect: $(k - 5)x + 3k = 3x + 2m$.

2. Infinitely many solutions require the x -terms to match and the constants to match: $k - 5 = 3 \Rightarrow k = 8$, and $3k = 2m \Rightarrow 24 = 2m \Rightarrow m = 12$.

3. $m - k = 12 - 8 = 4$. (The options 8, 12 and 20 are k , m and $k + m$.)

Answer: (A)

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

The equation $|x + 6| - 7 = \frac{x}{3} + c$, where c is a constant, has exactly one solution x . What is the value of $2c + x$?

(A) -11

(B) -6

(C) -5

(D) -16 ✓

1. The graph of $y = |x + 6| - 7$ is a V with vertex $(-6, -7)$ and arms of slope ± 1 . The line $y = \frac{x}{3} + c$ is less steep than either arm.

2. Such a line meets the V twice if it passes above the vertex, never if it passes below, and exactly once only if it passes through the vertex.

3. Through $(-6, -7)$: $-7 = \frac{-6}{3} + c \Rightarrow c = -5$, and the solution is $x = -6$.

4. $2c + x = -10 - 6 = -16$.

5. (Algebra check: for $x \geq -6$, $x = \frac{3(c+1)}{2}$ is valid when $c \geq -5$; for $x < -6$, $x = -\frac{3(c+13)}{4}$ is valid when $c > -5$. Exactly one solution occurs only at $c = -5$.)

Answer: (D)

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers $k \neq 3$ and $x \neq 0$:

$$\frac{4(k-3)}{x(k-3)} + \frac{2}{x} - \frac{3}{4} = \frac{3}{8}$$

What is the value of x ?

1. Cancel the common factor $(k-3)$: $\frac{4}{x} + \frac{2}{x} - \frac{3}{4} = \frac{3}{8}$.
2. Combine: $\frac{6}{x} = \frac{3}{8} + \frac{3}{4} = \frac{9}{8}$.
3. Solve for $\frac{1}{x}$: $\frac{1}{x} = \frac{9}{8 \cdot 6} = \frac{3}{16}$.
4. Flip at the end: $x = \frac{16}{3}$.

Answer: $\frac{16}{3}$ **34.**

SAT CHALLENGE · DESMOS

For an integer k , the equation

$$9 - 3|2x - 8| = 1 - \frac{5k}{2}$$

has no real solution x . What is the greatest possible value of k ?

1. The expression $9 - 3|2x - 8|$ is largest when $2x - 8 = 0$, where it equals 9. So the left side is at most 9.
2. There is no solution exactly when the right side is greater than 9: $1 - \frac{5k}{2} > 9$.
3. $-\frac{5k}{2} > 8 \Rightarrow k < -\frac{16}{5}$ (dividing by $-\frac{5}{2}$ reverses the inequality).
4. The integers satisfying $k < -3.2$ are $\dots, -5, -4$. The greatest is -4 .
5. (Check: $k = -4$ gives right side $11 > 9$, no solution; $k = -3$ gives $8.5 \leq 9$, and $|2x - 8| = \frac{1}{6}$ has solutions.)

Answer: -4

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{3x+2}{4} - \frac{x-7}{6} = \frac{5x-1}{12} + 2$$

If x is the solution to the equation above, which of the following intervals contains the value of $x^2 + 4x - 1$?

(A) Greater than 50

(B) Between 0 and 5

(C) Between 5 and 10 ✓

(D) Between 10 and 50

1. Multiply every term by the LCM 12: $3(3x+2) - 2(x-7) = (5x-1) + 24$.

2. Distribute (the minus sign applies to both terms of $x-7$): $9x+6-2x+14=5x+23$, so $7x+20=5x+23$.

3. $2x=3 \Rightarrow x=\frac{3}{2}$.

4. $x^2+4x-1=\frac{9}{4}+6-1=\frac{29}{4}=7.25$, which lies between 5 and 10.

5. (A sign slip, $-2x-14$, gives $x=\frac{31}{2}$ and a value above 50.)

Answer: (C)

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{2}{a} - \frac{3}{b} = \frac{5}{c}$$

In the equation above, a , b and c are nonzero constants and $2c \neq 5a$. Which of the following expresses b in terms of a and c ?

(A) $b = \frac{3ac}{2c+5a}$ **(B) $b = \frac{3ac}{2c-5a}$ ✓**(C) $b = \frac{3ac}{5a-2c}$ (D) $b = \frac{2c-5a}{3ac}$

1. Isolate the b -term: $\frac{3}{b} = \frac{2}{a} - \frac{5}{c}$.

2. Common denominator on the right: $\frac{2}{a} - \frac{5}{c} = \frac{2c-5a}{ac}$.

3. So $\frac{3}{b} = \frac{2c-5a}{ac}$. Take reciprocals and multiply by 3: $b = \frac{3ac}{2c-5a}$.

4. Check with $a=1$, $c=5$: $2 - \frac{3}{b} = 1 \Rightarrow b=3$, and $\frac{3(5)}{10-5} = 3$. The reversed-sign option gives -3 , the inverted option gives $\frac{1}{3}$, and $\frac{15}{15} = 1$ for the last.

Answer: (B)

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

Study room A has x students and study room B has 72 students. One-third of the students in room A move to room B. Then one-fourth of the students who are now in room B move to room A. After both moves, the ratio of the number of students in room A to the number in room B is 5 to 7. How many students were in room A originally?

(A) 45

(B) 63

(C) 72

(D) 36 ✓

1. After the first move: A has $\frac{2}{3}x$ and B has $72 + \frac{x}{3}$.
2. Second move: one-fourth of B's students, $\frac{1}{4}\left(72 + \frac{x}{3}\right) = 18 + \frac{x}{12}$, go to A. A now has $\frac{2}{3}x + 18 + \frac{x}{12} = \frac{3}{4}x + 18$. B keeps $\frac{3}{4}\left(72 + \frac{x}{3}\right) = 54 + \frac{x}{4}$.
3. Ratio 5 : 7: $7\left(\frac{3}{4}x + 18\right) = 5\left(54 + \frac{x}{4}\right) \Rightarrow 5.25x + 126 = 270 + 1.25x \Rightarrow 4x = 144 \Rightarrow x = 36$.
4. Check: A: $36 \rightarrow 24 \rightarrow 24 + 21 = 45$; B: $72 \rightarrow 84 \rightarrow 63$; $45 : 63 = 5 : 7$.
5. The values 45 and 63 are the final counts, and 72 is the original count of room B.

Answer: (D)**38.**

SAT CHALLENGE · NO CALCULATOR NEEDED

If $2x + 3y = 8$, which of the following is equivalent to $2x^2 + 3xy - 4x - 6y$?

(A) $8x - 16$ ✓(B) $8x - 8$ (C) $16 - 8x$ (D) $8x - 4y$

1. Group the terms: $(2x^2 + 3xy) - (4x + 6y) = x(2x + 3y) - 2(2x + 3y)$.
2. Factor out the common binomial: $(2x + 3y)(x - 2)$.
3. Substitute $2x + 3y = 8$: $8(x - 2) = 8x - 16$.
4. (Check with $x = 1$, $y = 2$, which satisfy $2x + 3y = 8$: the expression is $2 + 6 - 4 - 12 = -8$ and $8(1) - 16 = -8$.)

Answer: (A)

39.

SAT CHALLENGE · DESMOS

$$|2x - k| + k = x + 6$$

In the equation above, k is a constant. The equation has exactly one solution. What is the value of k ?

1. Graph $y = |2x - k| + k$: a V with vertex $\left(\frac{k}{2}, k\right)$ and arms of slope ± 2 . The line $y = x + 6$ has slope 1, shallower than both arms.
2. A shallower line meets such a V twice if it passes above the vertex, never if it passes below, and exactly once only if it passes through the vertex.
3. Put the vertex on the line: $k = \frac{k}{2} + 6 \Rightarrow \frac{k}{2} = 6 \Rightarrow k = 12$.
4. Check: $|2x - 12| + 12 = x + 6$ gives $x = 6$ from the right branch, and the left branch gives $x = 6$ as well (the vertex), so there is one solution.

Answer: 12

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$k(3x - 2) - (x - m) = 8x + 10$$

In the equation above, k and m are constants. The equation is true for every real value of x . What is the value of $k + m$?

1. Expand the left side: $3kx - 2k - x + m = (3k - 1)x + (m - 2k)$.
2. True for every x means the x -coefficients match and the constants match: $3k - 1 = 8$ and $m - 2k = 10$.
3. $k = 3$, then $m = 10 + 2(3) = 16$.
4. $k + m = 3 + 16 = 19$.

Answer: 19